

Walmart Data Analysis

1. Distribution of Transaction by Payment Method

Analyze how customers prefer to pay by counting transactions for each payment method

payment_method	count
Credit card	4256
Ewallet	3881
Cash	1832

2. Number of Branches Operating

Identify the total number of unique Walmart branches represented in the dataset

branches_count
100

3. Minimum Quantity Purchased

Find the smallest quantity of items purchased in a single transaction.

Minimum Quantity
1.0

4. Payment Method Performance Analysis

Evaluate payment method usage by measuring both transaction volume and total quantity of products sold through each payment method.

payment_method	no_payments	no_qty_sold
Credit card	4256	9567.0
Ewallet	3881	8932.0
Cash	1832	4984.0

5. Highest Rated Product Category by Branch

Identify the best-performing product category in each branch based on average customer ratings.

branch	category	avg_rating	rank
WALM001	Electronic accessories	7.45	1
WALM002	Food and beverages	8.25	1
WALM003	Sports and travel	7.5	1
WALM004	Food and beverages	9.3	1
WALM005	Health and beauty	8.366666666666667	1

6. Total Quantity Sold by Payment Method

Measure product sales volume associated with each payment method.

payment_method	no_qty_sold
Credit card	9567.0
Ewallet	8932.0
Cash	4984.0

7. Profitability Analysis by Product Category

Calculate total revenue and estimated profit for each category to identify the most product segments.

category	total_revenue	profit
Fashion accessories	489480.89999999997	192314.89320000037
Electronic accessories	78175.02999999998	30772.489499999978
Health and beauty	46851.17999999998	18671.7345
Food and beverages	53471.280000000006	21552.862200000003
Sports and travel	52497.930000000002	20613.808199999996
Home and lifestyle	489250.06	192213.63809999999

8. Revenue Decline Analysis Between 2022 and 2023

Identify the top 5 branches experiencing the largest percentage decline in revenue from 2022 to 2023 helping management detect underperforming locations.

branch	last_year_revenue	cr_year_revenue	rev_dec_ratio
WALM045	1731.0	647.0	62.62

branch	last_year_revenue	cr_year_revenue	rev_dec_ratio
WALM047	2581.0	1069.0	58.58
WALM098	2446.0	1030.0	57.89
WALM033	2099.0	931.0	55.65
WALM081	1723.0	850.0	50.67